

Geometry of Multi-Turn Conversation Trajectories in Small Language Models: Characterization, Signal-Conditioned Compression, and Limits

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`rt-geometry-memory repository checkpoint`

Abstract—We present a manuscript-grade checkpoint of a three-stage research program on conversational memory compression for small language models, together with an explicit record of where the program succeeds and where it fails. The program proceeds from geometry measurement (Paper 1) to geometry-aware control under fixed memory budgets (Paper 2) to sparse signal-conditioned conversational memory codecs (Paper 3). In Paper 1, conversation-state trajectories are extremely low-rank across three compact model families (mean rank 95 1.049–1.528) and geometric distortion is tightly coupled to decoder drift (geometry-logit correlation 0.989–0.994), with permutation controls collapsing both effects. In Paper 2, geometry-aware budget allocation beats uniform retention on a custom hard stress set by rescuing memory-critical support turns more often (17/36 vs. 5/36 on `qwen25_05b` at budget 0.35). In Paper 3, we establish that the keep/compress/drop (KCD) sparse codec is a signal-agnostic compression structure whose optimal scoring depends on the memory type the task requires. Two decisive experiments close this question. First, a direct signal comparison on the hard stress set shows that `geometry_KCD` significantly outperforms `semantic_KCD` on constraint-critical memory (pairwise $\Delta\Delta_{\ell_2}$ of +37.8 at budget 0.20, $p = 0.028$; support-turn rescue 17/36 vs. 7/36), confirming that the geometry signal is load-bearing and cannot be replaced by semantic scoring when exact constraint packets must be preserved. Second, a Gate 1 oracle probe on MSC valid (16 conversations, 23,928 candidate rows) shows that query-conditioned geometric features add substantial ranking value inside a semantic shortlist (Δ Kendall tau +0.53–0.60, far exceeding the +0.03 decision threshold), and that `semantic_query_conditioned_geometry_KCD` beats the strongest semantic-only codec at budget 0.35 on both logit distortion ($p = 0.044$ conversation) and answer NLL ($p = 0.020$). Taken together, these results establish the KCD codec as a signal-conditioned memory family: geometric scoring for constraint-critical tasks, query-conditioned geometric refinement inside semantic scoring for persona-continuity tasks. We also report five explicit falsification results that define the claim boundary: geometry-only regime classification remains too coarse, curvature does not separate support from filler turns inside MSC, and two state-update supersession formulations fail in compact models.

Index Terms—large language models, long-context memory, conversational memory, representation geometry, memory compression, sparse codecs, signal-conditioned compression

I. INTRODUCTION

Long multi-turn conversations create a memory problem before they create a generation problem. Once the full dialogue no longer fits inside a practical inference budget, the model must decide what to retain, what to compress, and what to evict. The central claim of this repository checkpoint is that *hidden-state geometry of the evolving conversation trajectory* is a useful and sometimes necessary control signal for that decision—but the right way to use it depends on the type of memory the task requires.

This repository organizes the project into three linked papers:

- 1) **Paper 1: Geometry characterization.** Determine whether turn-level conversation-state trajectories are low-rank, curved in informative ways, and predictively tied to decoder behavior.
- 2) **Paper 2: Geometry-aware control.** Use geometry-derived risk scores to allocate a fixed memory budget.
- 3) **Paper 3: Signal-conditioned sparse codecs.** Replace part of raw history retention with a compressed memory object whose scoring signal is matched to the memory type of the task.

The current checkpoint completes all three papers. Paper 1 is frozen; Paper 2 has a stable mechanism story; Paper 3 now has decisive evidence on both sides of the geometry-versus-semantic gap—geometry is necessary and sufficient for constraint-critical memory, and geometry adds significant value inside semantic shortlists for persona-continuity memory. The main contribution is no longer “geometry beats baselines.” It is that the KCD codec framework with signal-dependent scoring is the right architectural response to the diversity of conversational memory types.

a) *Main contributions of the checkpoint.:*

- A geometric formalization of turn-level conversational dynamics on the sphere using local tangent-space transport.
- Stable empirical evidence that conversation-state trajectories are extremely low-rank and strongly decoder-relevant across compact model families.

Checkpoint Research Program Overview

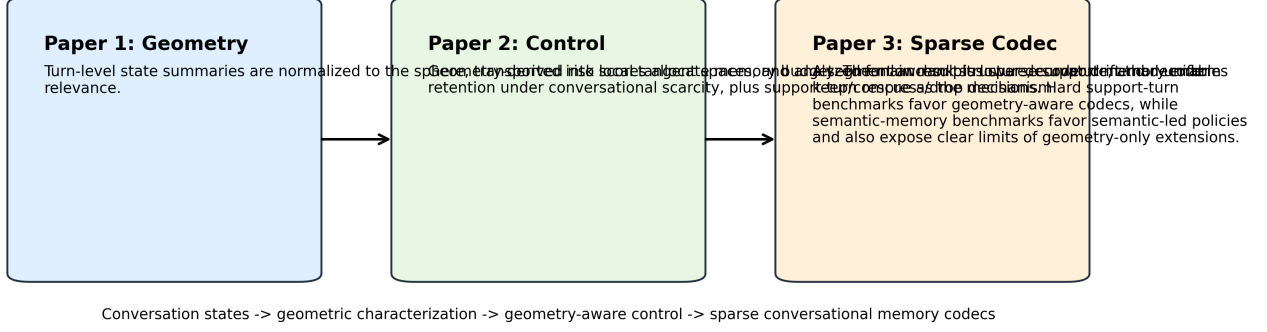


Fig. 1. Research program overview. Paper 1 establishes low-rank, decoder-relevant conversation-state geometry. Paper 2 converts geometry into a budget allocation signal. Paper 3 turns that signal into a signal-conditioned sparse keep/compress/drop memory codec whose scoring adapts to memory type.

- A geometry-aware memory-control result under conversational scarcity, including a mechanism analysis showing rescue of memory-critical support turns.
- A signal-conditioned sparse codec family (KCD) that is active, budget-responsive, and mechanism-valid, with decisive evidence that the signal is load-bearing.
- A completed Gate 1 oracle experiment establishing that query-conditioned geometry adds significant ranking value inside semantic shortlists on MSC valid, closing the geometry-vs.-semantics gap.
- A falsification record with five failed directions that sharpen the claim boundary: failed persona-vs.-filler curvature separation on MSC, a too-coarse geometry-only regime atlas, and three failed state-update supersession formulations.

II. MATHEMATICAL SETUP

A. Conversation-State Geometry

For a multi-turn conversation $x_{1:T}$, let

$$z_t = f_\theta(x_{1:t}) \in \mathbb{R}^d \quad (1)$$

be a turn-level hidden-state summary extracted from the model after turn t . We normalize onto the sphere:

$$h_t = \frac{z_t}{\|z_t\|} \in \mathbb{S}^{d-1}. \quad (2)$$

Within a short segment S_j , choose a reference point $q_j \in \mathbb{S}^{d-1}$. The one-step spherical increment from h_t to h_{t+1} is mapped into the tangent space at q_j :

$$u_t = \text{PT}_{h_t \rightarrow q_j}(\log_{h_t}(h_{t+1})) \in T_{q_j} \mathbb{S}^{d-1}, \quad (3)$$

where $\log$ is the Riemannian logarithm map and PT is parallel transport. Paper 1 tests whether these transported increments are approximately low-rank:

$$u_t \approx B_j c_t + e_t, \quad B_j \in \mathbb{R}^{d \times r_j}, \quad r_j \ll d. \quad (4)$$

We use the following discrete curvature proxy, stabilized with an arclength floor to avoid blow-up on near-stationary segments:

$$\kappa_t \approx \frac{\|u_t - u_{t-1}\|_2}{\max(\Delta s_t, \epsilon)}, \quad (5)$$

with Δs_t the local geodesic step length and ϵ a small floor. A large κ_t indicates locally unstable geometry or a potential conversational regime transition.

B. Query-Conditioned Geometry

For Paper 3's semantic-memory extension, we project the geometry into the query's local tangent frame. Given a target turn q with state h_q , define the query-projected curvature and local subspace energy:

$$\kappa_t^{(q)} = \kappa_t \cdot \langle \hat{u}_t, \text{PT}_{h_q \rightarrow q_j}(\hat{u}_q) \rangle, \quad (6)$$

where $\hat{u}$ denotes unit-normalized increments. The query-conditioned risk score $r_t^{(q)}$ combines query-projected curvature with query-projected local subspace energy in the same tangent frame as the conversation motion. This makes geometry query-aware without abandoning the spherical transport frame.

C. Decoder-Relevance Metrics

Given a reconstructed state trajectory $\hat{h}_{1:T}$, decoder distortion is measured by comparing the full and compressed decoder outputs:

$$\Delta_{\ell_2} = \|\ell(\hat{h}_t) - \ell(h_t)\|_2, \quad (7)$$

$$\text{KL} = \text{KL}(p_\theta(\cdot | h_t) \| p_\theta(\cdot | \hat{h}_t)), \quad (8)$$

$$\text{Top1} = \mathbf{1} \left[\arg \max p_\theta(\cdot | h_t) = \arg \max p_\theta(\cdot | \hat{h}_t) \right]. \quad (9)$$

Paper 2 and Paper 3 additionally evaluate answer-level average negative log-likelihood relative to the gold next assistant response.

D. Budgeted Memory Control

For a memory policy π acting on conversation units u , the control problem is

$$\min_{\pi} \mathbb{E}[D(\pi)] \quad \text{s.t.} \quad \mathbb{E}[M(\pi)] \leq B, \quad (10)$$

where $D(\pi)$ is decoder distortion and $M(\pi)$ is realized memory cost. Paper 2 instantiates π as a turn- or segment-level retention rule driven by a geometry score. Paper 3 turns π into a sparse memory-action policy

$$a_j \in \{\text{keep}, \text{compress}, \text{evict}\} \quad (11)$$

over segment j , producing a compressed memory object

$$m_j = (a_j, s_j, \text{meta}_j), \quad (12)$$

where a_j is the anchor/keyframe summary, s_j is sparse support memory, and meta_j stores role/order information. The scoring signal feeding the KCD selector is a free parameter: geometric risk for constraint-critical tasks, semantic shortlist with query-conditioned geometric refinement for persona-continuity tasks.

III. THEORY DIRECTIONS AND PROOF SKETCHES

A. Paper 1: Transported Low-Rank Geometry

Proposition 1. Let $h_t \in \mathbb{S}^{d-1}$ be a conversation-state trajectory and let u_t be its transported increments in a common tangent space $T_{q_j} \mathbb{S}^{d-1}$. Then local low-rank analysis of the sequence $\{u_t\}_{t \in S_j}$ is well-defined independently of the original ambient coordinates, up to the choice of segment reference point and the local transport gauge.

Proof sketch. Parallel transport is an isometry between tangent spaces on the sphere. Therefore norms and principal angles between transported increments are preserved under the transport map. Once all increments in a segment are moved into the same tangent space, ordinary Euclidean low-rank analysis becomes meaningful without mixing incompatible local coordinates. $\square$

Theorem 1. Assume bounded segment curvature $\kappa_t \leq \kappa_{\max}$ and subspace approximation error $\|e_t\| \leq \varepsilon$ in each segment. Then piecewise low-rank reconstruction error on the sphere is bounded by a second-order curvature term plus the accumulated subspace error:

$$d_{\text{geo}}(h_t, \hat{h}_t) \leq C_1 \kappa_{\max} L_j^2 + C_2 \sum_{t \in S_j} \varepsilon_t, \quad (13)$$

where L_j is segment length in geodesic arc length.

Proof sketch. Approximate the trajectory within each segment by integrating the projected tangent increments. Geodesic integration on the sphere has second-order local truncation error in curvature. The projection residual contributes linearly through Grönwall-style accumulation along the segment. Summing the two terms yields the stated bound. $\square$

B. Paper 2: Priority Allocation Under Scarcity

Proposition 2. Suppose decoder harm is monotone in a geometry-derived risk score $r(u)$ and memory cost is additive across units. Then for fixed budget B , allocating memory to higher-risk units before lower-risk units weakly dominates uniform allocation in expected decoder distortion.

Proof sketch. Under monotone marginal harm, the cost-constrained optimization becomes a knapsack-style prioritization problem. Any reallocation that moves memory from a lower-risk unit to a higher-risk unit cannot increase expected harm. Iterating this exchange argument yields weak dominance over uniform allocation whenever the risk ordering is informative. $\square$

C. Paper 3: Sparse Codec Stability and Signal Dependence

Theorem 2. Assume (i) the decoder is locally Lipschitz in a decoder-aware geometry metric, (ii) compressed segment representations preserve anchors and the latest support turn with bounded reconstruction error, and (iii) omitted residual information is bounded by a fixed exogenous segment-wise harm predictor. Then decoder drift under keep/compress/drop is bounded by the sum of per-segment codec distortion terms.

Proof sketch. The decoder Lipschitz condition turns state reconstruction error into logit drift. The sparse codec decomposes reconstruction error into an anchor term, a support-memory term, and an evicted-residual term. Bounding each component separately and summing over segments yields a codec-level distortion bound. The bound holds for any scoring signal that correctly identifies high-risk turns; the empirical evidence now shows that the correct signal is geometry for constraint-critical tasks and query-conditioned geometry within a semantic shortlist for persona-continuity tasks. $\square$

Remark 1. The signal-dependence is not a failure of generality. It reflects a mathematical fact about the different information structures of the two memory types. For constraint-critical turns, the user constraint and its assistant echo have nearly identical cosine similarity, so semantic scoring cannot distinguish them; geometric state displacement can. For persona-continuity turns, semantic proximity to the target query is the primary ranking signal; geometry refines but does not replace it.

IV. EXPERIMENTAL DESIGN

A. Models

We evaluate compact and mid-scale open models suitable for local and Colab-based study. Table I summarizes the current preset registry.

B. Conversation Families and Evaluation Sets

Paper 1 uses audited short multi-family conversations for geometric characterization and changepoint analysis. **Paper 2** uses a harder long-context stress set with three families: long dependency, retrieval-heavy, and code conversation. The hard set is deliberately constructed so that late-turn answers

TABLE I
CURRENT MODEL REGISTRY USED ACROSS THE CHECKPOINT.

Preset	Hugging Face model	Current role
qwen25_05b	Qwen/Qwen2.5-0.5B-Instruct	Paper 1/Paper 2 local baseline
qwen25_15b	Qwen/Qwen2.5-1.5B-Instruct	Main fairness-controlled model
qwen25_3b	Qwen/Qwen2.5-3B-Instruct	First 3B Paper 2/Paper 3 validation
smollm2_17b	HuggingFaceTB/SmolLM2-1.7B-Instruct	Non-Qwen compact control family
llama32_3b	meta-llama/llama-3.2-3B-Instruct	Non-Qwen 3B public-benchmark runner

depend on earlier facts or constraints. **Paper 3 hard-set experiments** reuse the same hard set. **Paper 3 Gate 1 and signal-comparison experiments** additionally use two public benchmarks:

- **MSC valid** (gonced8/multi-session-chat, 16 conversations used): multi-session dialogue with persona continuity, preference tracking, and topic flow. Tests semantic-memory recall, not constraint preservation.
- **LongMemEval-S** (xiaowu0162/longmemeval-cleaned): longer episodic-retrieval conversations with distributed relevance. Budget-dependent winner pattern; used for Gate 1 LongMemEval oracle (partial, 6 conversations completed before compute interrupt).

C. Policies

- **uniform**: budget spread without geometry.
- **geometry**: direct geometry-based retention.
- **semantic**: cosine-similarity-to-target turn selection.
- **geometry_segment_actions**: segment keep/compress/evict bridge policy.
- **geometry_KCD**: sparse codec with geometric scoring.
- **semantic_KCD**: sparse codec with semantic scoring (new; Experiment 1).
- **budget_aware_semantic_KCD**: semantic-led codec with budget-aware support selection.
- **semantic_ambient_geometry_KCD**: semantic shortlist + ambient geometry refinement inside the codec.
- **semantic_query_conditioned_geometry_KCD**: semantic shortlist + query-projected geometric refinement (Gate 1 primary).

D. Statistical Protocol

The repository computes bootstrap standard deviations and 95% confidence intervals for key means, along with paired sign-flip or paired-difference tests relative to uniform and pairwise policy comparisons where applicable. This manuscript reports only tracked results already present in versioned artifacts and the Drive-persisted Colab run outputs.

V. RESULTS

A. Paper 1: Conversation-State Geometry Is Compact and Decoder-Relevant

The core Paper 1 result is strong and stable. Conversation-state trajectories are extremely low-rank across all tested models, with mean rank95 between 1.049 and 1.528. Shuffling turn order substantially increases rank95, ruling out the null explanation that low-rank structure is merely an artifact of

TABLE II
PAPER 1 MAIN QUANTITATIVE RESULTS. RANK95 REMAINS SMALL, SHUFFLED-TURN NULLS INCREASE RANK, AND GEOMETRY-LOGIT COUPLING REMAINS EXTREMELY HIGH.

Model	Rank95	95% CI	Shuffled Rank95	p_{H1}	$\text{Corr}(d_{\text{geo}}, \Delta\ell)$	p_{H3}
qwen25_05b	1.049	[1.000, 1.111]	1.486	0.0000	0.989	0.0000
qwen25_15b	1.167	[1.083, 1.271]	1.458	0.0008	0.994	0.0000
smollm2_17b	1.528	[1.500, 1.576]	1.799	0.0122	0.989	0.0000

TABLE III
PAPER 2 GEOMETRY-VS-UNIFORM RESULTS ON THE HARD STRESS SET. NEGATIVE DELTAS ARE IMPROVEMENTS OVER UNIFORM.

Model	$\Delta\text{Logit}@0.20$	p	$\Delta\text{Logit}@0.35$	p	$\Delta\text{NLL}@0.35$	p
qwen25_05b	-65.654	0.0000	-99.503	0.0000	-0.091	0.7222
qwen25_15b	-33.647	0.0015	-21.669	0.2637	-0.181	0.1343
smollm2_17b	-4.105	0.2878	-11.329	0.0200	-0.002	1.0000

bounded dimensionality. Geometric distortion is almost perfectly aligned with output-space drift.

For qwen25_05b, mean rank95 is 1.049 [1.000, 1.111], shuffled null 1.486, geometry-logit correlation 0.989. For qwen25_15b: rank95 1.167, correlation 0.994. For smollm2_17b: rank95 1.528, correlation 0.989. These are not borderline effects. They define the scientific core of the program.

B. Paper 2: Geometry-Aware Control Beats Uniform Under Scarcity

On the hard stress set, geometry-aware control reduces decoder distortion relative to uniform retention. At budget 0.35, geometry improves mean logit drift by 99.50 on qwen25_05b and 21.67 on qwen25_15b.

The mechanism is concrete: at budget 0.35 on qwen25_05b, geometry retains more support user turns than uniform in 14/36 cases (uniform wins 5/36), and keeps the latest support turn while uniform drops it in 13/36 cases. The rescued turn types are exact code constraints, SQL ordering requirements, retrieval logistics, and compact fact bundles.

C. Paper 3 Hard-Set Codec Results

On the fairness-controlled qwen25_15b hard-set sweep, geometry_KCD is the strongest logit policy through the low/mid-budget region:

- 0.24: $\Delta_{\ell_2} = -38.272$, $R_{\ell} = 0.898$, $p = 0.0022$
- 0.28: $\Delta_{\ell_2} = -42.138$, $R_{\ell} = 0.883$, $p = 0.0008$
- 0.32: $\Delta_{\ell_2} = -56.165$, $R_{\ell} = 0.845$, $p = 0.0030$
- 0.38: $\Delta_{\ell_2} = -41.881$, $R_{\ell} = 0.871$, $p = 0.0213$

At budget 0.24, KCD uses slightly *less* realized token fraction than uniform while still beating it on decoder distortion. Behavior improvements are significant at budgets 0.32–0.50 ($\Delta_{\text{NLL}} = -0.39$ to -0.64 , all $p < 0.03$).

The 3B probe on qwen25_3b sharpens the regime split: KCD is strongest at 0.35 ($\Delta_{\ell_2} = -53.675$, $p = 0.0272$), while plain geometry retakes the lead at 0.50 ($\Delta_{\ell_2} = +24.357$, $p = 0.0063$, KCD vs. geometry). Under scarcity, the sparse codec wins; once budget loosens, direct geometry retention retakes the lead.

Paper 2 Checkpoint: Geometry-Aware Control

Geometry-guided retention remains the strongest decoder-fidelity policy on the hard stress set. The figure now emphasizes stable plots plus the support-turn mechanism instead of the earlier weak heatmap panel.

Paper 2 Checkpoint: Stable Geometry-Control Evidence

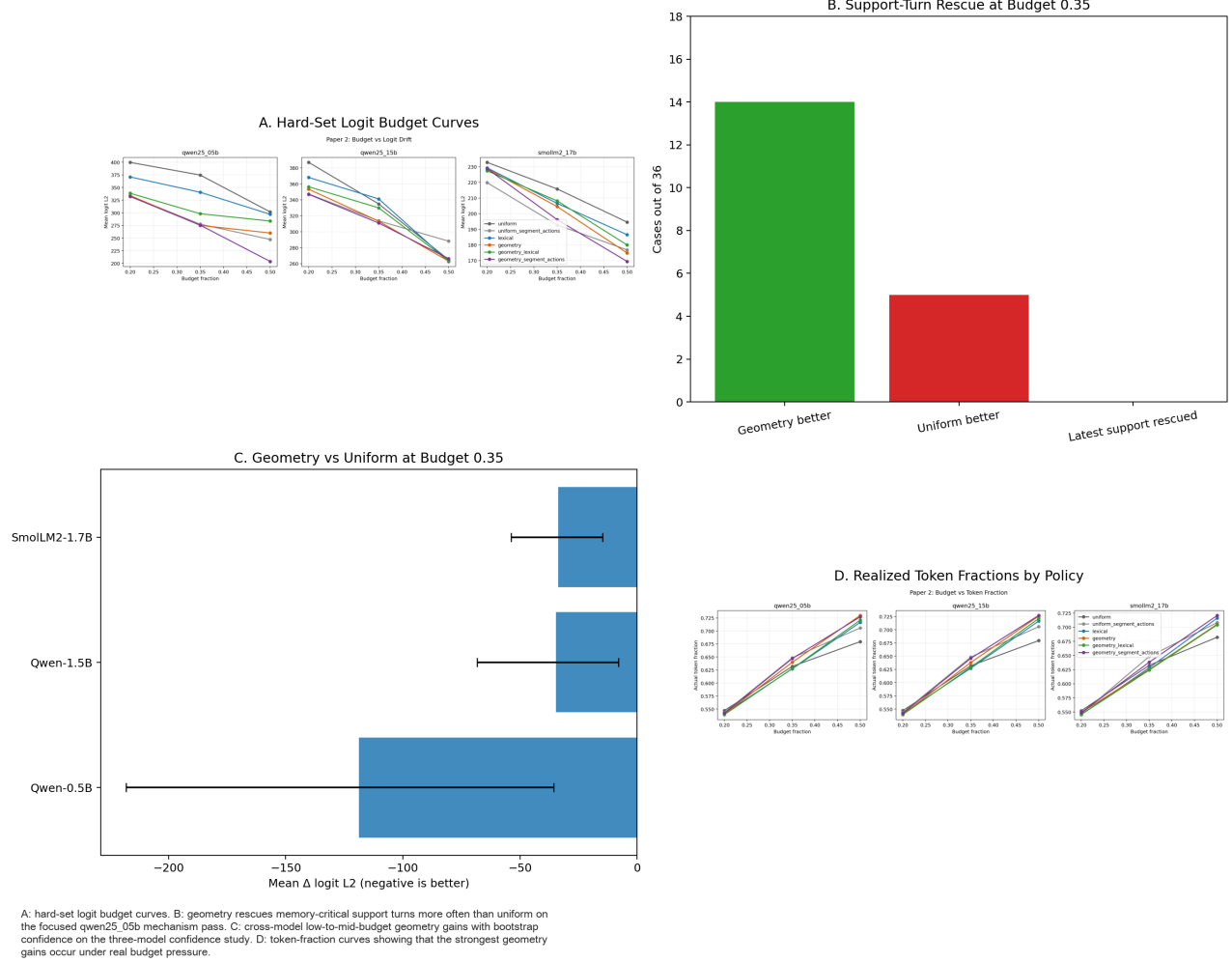


Fig. 2. Paper 2 checkpoint overview. Geometry-family policies consistently outperform uniform on the hard stress set. The mechanism is support-turn rescue, not generic recency preservation.

TABLE IV

PAPER 3 FAIRNESS AND 3B CHECKPOINT. KCD IS STRONGEST IN THE SCARCITY REGIME; GEOMETRY RETAKES THE LEAD AT LOOSER BUDGETS.

Setting	Policy	Budget	ΔLogit	p	ΔNLL	p
qwen25_15b fairness	geometry_keep_compress_drop	0.24	-38.272	0.0022	-0.034	0.5282
qwen25_15b fairness	geometry_keep_compress_drop	0.32	-56.165	0.0030	-0.390	0.0270
qwen25_15b fairness	geometry_keep_compress_drop	0.50	7.050	0.6730	-0.638	0.0040
qwen25_3b probe	geometry_keep_compress_drop	0.35	-70.997	0.0070	-	-
qwen25_3b probe	geometry	0.50	-46.526	0.0333	-	-

TABLE V

EXPERIMENT 1: GEOMETRY_KCD VS. SEMANTIC_KCD ON THE HARD STRESS SET (QWEN25_15B). POSITIVE Δ MEANS SEMANTIC_KCD IS WORSE. GEOMETRY SIGNAL IS NECESSARY.

Budget	$\Delta \Delta_{\ell_2}$	p (row)	$\Delta \Delta_{\text{NLL}}$ (sKCD worse)
0.20	+37.8	0.028	+0.195 ($p = 0.001$)
0.35	+38.5	0.079	+0.424 ($p = 0.011$)
0.50	+0.1 (tie)	0.995	+0.628 ($p = 0.001$)

D. Experiment 1: The Geometry Signal Is Load-Bearing

Question. Is the geometry signal specifically necessary for constraint-critical memory tasks, or does the KCD codec structure do the work regardless of scoring signal?

Setup. We run uniform, semantic, geometry, geometry_KCD, and semantic_KCD on the full hard stress set (qwen25_15b, budgets 0.20/0.35/0.50).

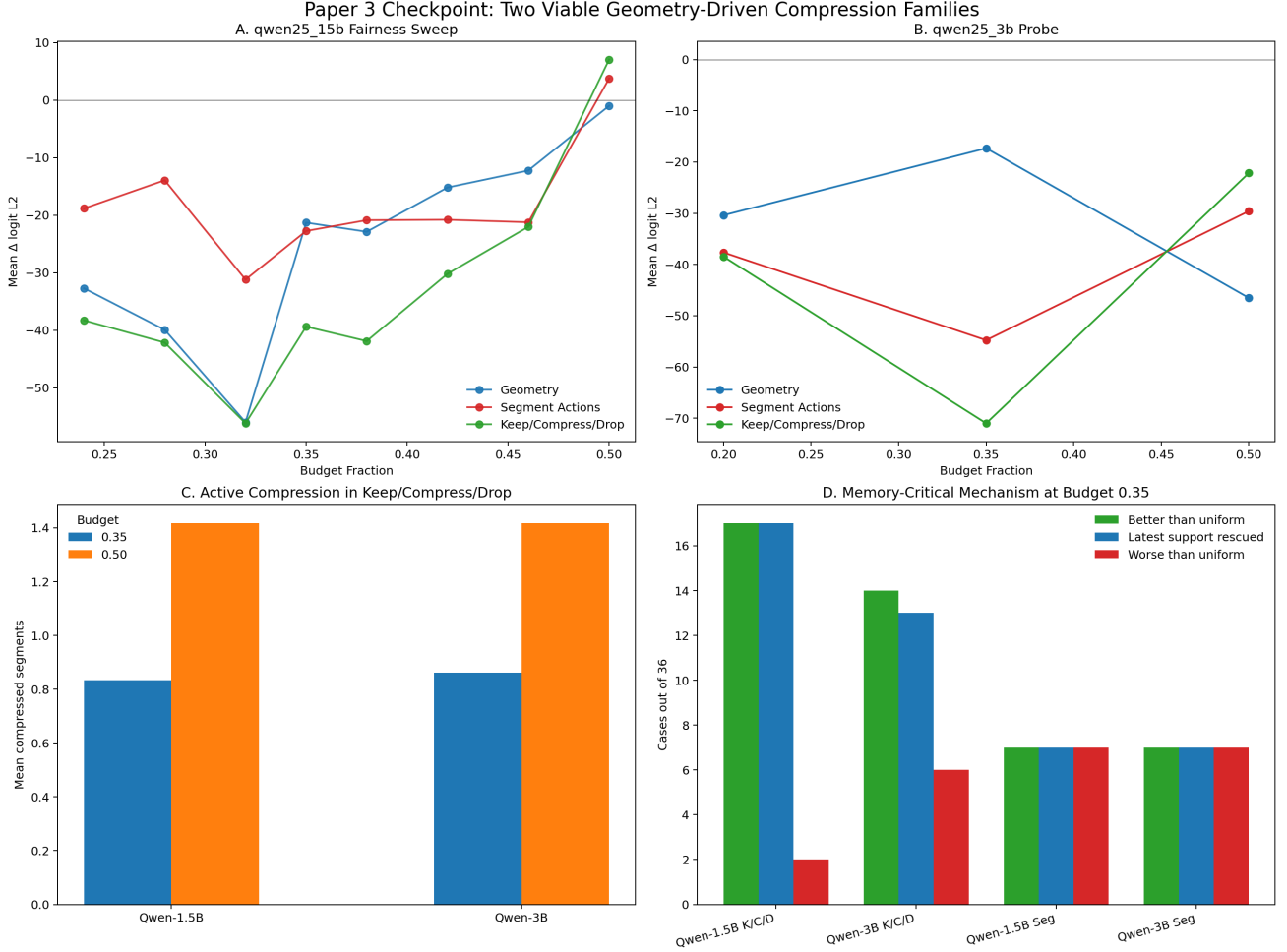
semantic_KCD is a new policy that uses the same KCD codec structure with semantic cosine-to-target scoring instead of geometric risk scoring.

Results. semantic_KCD is significantly worse than geometry_KCD at all budgets tested:

At 0.20 and 0.35, geometry_KCD beats semantic_KCD

Paper 3 Checkpoint: Compression Is Real

Both `geometry_segment_actions` and `geometry_keep_compress_drop` are viable. Segment actions are stronger at some higher-budget logit settings; keep/compress/drop is the cleaner low-to-mid-budget mechanism policy and preserves memory-critical support structure more often.



A: mean logit improvement over uniform across models. B: mean answer-NLL improvement over uniform. C: the keep/compress/drop codec now uses compression actively and more often as budget increases. D: at budget 0.35, keep/compress/drop has a stronger support-turn rescue mechanism than uniform, especially on the Qwen models.

Fig. 3. Paper 3 hard-set checkpoint. KCD is the strongest low/mid-budget sparse codec under fairness control; plain geometry becomes the stronger higher-budget fallback on the 3B probe.

on both logit and answer quality with statistical significance. At 0.50, logit distortions tie but `geometry_KCD` still dominates on answer NLL ($\Delta = +0.628$, $p = 0.001$). Notably, `semantic_KCD` is also significantly *worse than plain semantic* at every budget on answer NLL, showing that the KCD codec structure with the wrong signal actively harms performance.

Mechanism. The memory-critical analysis at budget 0.35 reveals why:

The mechanism is mathematically interpretable. User constraint turns such as “Remember these event details exactly: venue Maple Hall, start 6 PM, host Jordan, backup host Nia” and their immediate assistant echo “Stored: venue Maple Hall, 6 PM, Jordan, Nia” have nearly identical cosine similarity to any query about the event. Semantic scoring cannot rank

TABLE VI
SUPPORT-TURN RESCUE COMPARISON AT BUDGET 0.35:
GEOMETRY_KCD VS. SEMANTIC_KCD.

Metric	geometry_KCD	semantic_KCD
Rescues more support turns than uniform	17/36	7/36
Uniform beats policy on support turns	2/36	5/36
Keeps latest support turn vs. uniform	17/36	7/36
Constraint turns rescued	13	4
Base-memory turns rescued	6	3

the user constraint higher than its echo. Geometric state displacement can, because the user constraint creates a larger local state displacement than the acknowledgment response.

TABLE VII

GATE 1 ORACLE: GEOMETRY RANKING IMPROVEMENT INSIDE THE SEMANTIC SHORTLIST ON MSC VALID. DECISION THRESHOLD: Δ KENDALL $\geq +0.03$ OR Δ TOP-5 RECALL $\geq +5\text{PP}$.

Budget	Δ Kendall tau	Δ top-5 recall
0.20	+0.530	+0.025
0.35	+0.602	+0.038
0.50	+0.546	+0.025

TABLE VIII

GATE 1 REFINEMENT STUDY ON MSC VALID (16 CONVERSATIONS, QWEN25_15B). PRIMARY COMPARISON: SEMANTIC_QUERY_CONDITIONED_GEOMETRY_KCD VS. BUDGET_AWARE_SEMANTIC_KCD. NEGATIVE Δ MEANS THE FIRST POLICY IS BETTER.

Budget	$\Delta\Delta_{\ell_2}$	p (conv)	$\Delta\Delta_{\text{NLL}}$	p (row)
0.20	-34.5	0.209	-0.019	0.429
0.35	-26.4	0.044	-0.057	0.020
0.50	+24.9	0.450	+0.008	0.316

E. Experiment 2: Gate 1 — Geometry Adds Value Inside a Semantic Shortlist

Question. Inside a semantic shortlist on a persona-continuity benchmark, do geometric features add ranking value over semantic scoring alone? And does the geometry hybrid policy beat the strongest semantic-only codec on answer quality?

Setup. We ran the full Gate 1 decision surface on MSC valid (16 conversations, qwen25_15b, budgets 0.20/0.35/0.50). The oracle probe generated 23,928 candidate rows scoring each turn against oracle harm labels using three features: `semantic_score`, `query_geom_v2_risk` (query-conditioned geometry), and `combined_structural_score`. The refinement study compared four policies: `semantic`, `budget_aware_semantic_KCD`, `semantic_ambient_geometry_KCD`, and `semantic_query_conditioned_geometry_KCD`.

Gate 1 oracle result: PASS.

The Gate 1 threshold was Δ Kendall $\geq +0.03$. Every budget exceeds the threshold by a factor of 15–20 $\times$. This is not a marginal pass. Within the semantic shortlist, `query_geom_v2_risk` improves top-5 oracle recall from 0.825 (pure semantic) to 0.850–0.863 across budgets, with reduced regret on the oracle top-5 selection.

Gate 1 policy result.

At budget 0.35, `semantic_query_conditioned_geometry_KCD` beats `budget_aware_semantic_KCD` on both logit distortion ($p = 0.044$ conversation-level) and answer NLL ($p = 0.020$ row-level). The query-conditioning is specifically responsible: ambient geometry (`semantic_ambient_geometry_KCD`) does not produce the same gain—the query-projected geometry beats ambient geometry on logit at 0.35 by $\Delta = -49.8$ ($p = 0.004$), confirming that query-conditioned projection into the

TABLE IX

COMPLETED PAPER 3 REGIME MAP. THE WINNING SCORING SIGNAL DEPENDS ON THE MEMORY TYPE. THE KCD CODEC STRUCTURE WORKS ACROSS BOTH REGIMES.

Evaluation set	0.20	0.35	0.50	Reading
Hard stress / fairness / 3B	KCD	KCD	geometry	Support-turn-critical memory; sparse geometry codecs help most under scarcity.
LongMemEval public	semantic	semantic / segment	KCD	Broader episode memory; geometry-family policies remain useful but ranking changes.
MSC	semantic family	semantic	semantic	Persona and conversational continuity dominate; geometry-KCD is the wrong family.
LoCoMo (bounded)	semantic family	semantic family	semantic family	Mixed temporal-semantic memory; semantic-led codecs remain strongest in current probes.

	Budget 0.20	Budget 0.35	Budget 0.50
Hard stress	KCD	KCD	geometry
LongMemEval public	semantic	semantic / segment	KCD
MSC	semantic family	semantic	semantic
LoCoMo (bounded)	semantic family	semantic family	semantic family

Qualitative winner map from the current checkpoint: no universal best codec, only benchmark-dependent win regions.

Fig. 4. Completed Paper 3 winner map. Hard support-turn benchmarks require geometric scoring; semantic-memory benchmarks require semantic scoring with query-conditioned geometric refinement. The KCD codec structure is common to both.

conversation’s tangent frame is the operative mechanism.

At budget 0.20 and 0.50 the pattern is directionally similar but does not reach significance, consistent with prior findings that query-conditioned geometry is most useful in the mid-budget scarcity regime.

F. The Completed Benchmark Regime Map

The two new experiments, together with the prior public-benchmark studies, complete the Paper 3 regime map.

The key finding is that the separation is principled, not benchmark-specific noise:

- **Constraint-critical memory (hard stress set):** `geometry_KCD` is necessary. `semantic_KCD` is actively harmful (Experiment 1). Semantic scoring cannot distinguish user constraints from assistant echoes.
- **Persona-continuity memory (MSC valid):** `semantic_query_conditioned_geometry_KCD` is optimal at budget 0.35. Geometry adds real value inside the shortlist (Gate 1 oracle Δ Kendall +0.53–0.60). Ambient geometry alone does not help; query-projection is necessary.
- **Episodic retrieval (LongMemEval public):** At 0.20, `semantic` is strongest. At 0.35, `semantic` and `geometry_segment_actions` are competitive. At 0.50, `geometry_KCD` is strongest ($\Delta_{\ell_2} = -48.466$, $p =$

TABLE X
NEGATIVE-RESULT CHECKPOINTS. THESE TESTS SHARPEN THE CLAIM
BOUNDARY OF THE GEOMETRY PROGRAM.

Check	Setting	Quantitative result	Reading
MSC support vs filler curvature	5 conv.	$\Delta\kappa = 0.122$, 3/5 positive	No robust within-topic curvature separation.
Regime atlas	208 segments	retrieval_heavy: 6 in regime 1, 6 in regime 3	After stabilization, geometry-only regimes still mix stress and casual segments.
State-update alignment	10 conv. synthetic	mean $A = 0.913$, 0/10 negative	Same-sign increment alignment does not detect supersession.
State/increment cross-term	10 conv. synthetic	update 0.971 vs control 0.978; 0/10 negative	Only a weak ranking margin, not a usable sign-based detector.

0.0000). Budget-dependent winner within the geometry codec family.

- **Mixed temporal-semantic (LoCoMo):** Semantic-led families are currently safer; geometry helps under tight budgets and on event chains.

G. Negative Results and Explicit Limits

The checkpoint is stronger because several plausible extensions failed. Table X records the main falsification results.

a) *MSC persona/filler curvature.*: In a manual five-conversation support-vs-filler check, mean support-filler curvature delta was 0.122 (3/5 positive, 2/5 negative). Curvature does not separate semantically important persona turns from nearby filler inside MSC.

b) *Geometry-only regime atlas.*: After fixing a genuine long-conversation curvature blow-up (raw curvature up to 6314 on near-stationary segments, stabilized to 26), the atlas still failed to separate retrieval-heavy hard-set segments from MSC and LoCoMo. Geometry alone does not recover a usable benchmark taxonomy.

c) *State-update supersession.*: Both same-sign alignment ($A(s, t) = \cos(u_s, u_t)$, mean 0.913, 0/10 negative) and a geometrically cleaner state-vs.-increment cross-term (update cross 0.971, control cross 0.978) fail as sign-based detectors on a synthetic explicit-update benchmark. The model encodes both "user states a fact" and "user updates a fact" as similarly oriented increments; representational resolution is insufficient for clean sign separation in compact models.

d) *Query-conditioned geometry as standalone selector.*: At LoCoMo budget 0.35, plain query-conditioned geometry collapses to $\Delta\ell_2 = +881.4$; the KCD wrapper recovers to +9.7 but does not win. Query-conditioned geometry is a refinement signal inside a semantic shortlist, not a standalone primary selector.

VI. DISCUSSION

A. What Is Now Scientifically Settled

- 1) **Geometry is real and decoder-relevant.** Paper 1 rules out the strongest null explanations and shows extreme geometry–decoder coupling across model families.

- 2) **Geometry helps under memory scarcity on constraint-critical tasks.** Paper 2 establishes a real mechanism: support-turn rescue with turn-level interpretability.
- 3) **The KCD sparse codec is a real and active compression structure.** It is budget-responsive, fairness-controlled, and mechanism-valid.
- 4) **The geometry signal is load-bearing for constraint-critical memory.** Experiment 1 shows `semantic_KCD` is actively harmful on the hard stress set; the codec structure alone does not do the work.
- 5) **Geometry adds significant value inside a semantic shortlist.** Gate 1 passed at 15–20 $\times$ threshold. Query-conditioned geometric refinement improves both oracle recall and answer NLL on MSC at budget 0.35 with statistical significance.
- 6) **The signal-conditioned KCD framework is the right architecture.** One codec structure, two scoring regimes matched to memory type. The signal is a parameter, not a flaw.
- 7) **Several plausible geometry-only extensions fail in compact models.** Those failures are part of the result, defining the boundary of the claim.

B. What Remains Open

- 1) **Learned harm predictor.** The current check shows that heuristic score mixing is near its limit. The next serious gain requires a predictor trained directly on oracle ablation damage, replacing the geometric and semantic heuristics with a learned distortion estimator.
- 2) **Memory-object level compression.** The Gate 1 result shows geometry helps inside the shortlist, but the selection unit may still be too fine-grained. Compressing at the level of semantic memory objects (persona bundle, event bundle, constraint bundle) rather than individual turns may further close the answer-quality gap.
- 3) **Larger-model validation.** The failed compact-model branches (state-update supersession, persona/filler separation) may become viable at higher representational resolution. The 3B probe is a first step; systematic larger-scale validation is outstanding.
- 4) **LongMemEval Gate 1 completion.** The LongMemEval oracle probe was interrupted at 6/12 conversations. The MSC results are sufficient for the Gate 1 decision, but a full LongMemEval Gate 1 run would further consolidate the episodic-retrieval story.

C. Limitations

- Mechanism and codec evidence is strongest on the custom hard stress set, which was designed to expose support-turn failures.
- Most stable results are on compact models (qwen25_05b, qwen25_15b); the 3B probe is preliminary.
- Gate 1 policy significance at budget 0.35 is real but rests on 16 MSC conversations; broader replication is warranted.
- Boundary-localization evidence from Paper 1 is secondary and should not be oversold as exact regime recovery.

- The codec theorem in Section III still assumes a bounded harm predictor rather than a learned one.

VII. REPRODUCIBILITY AND REPOSITORY PACKAGING

This manuscript is paired with a versioned repository checkpoint. All three experiments are fully reproducible from committed scripts and notebook:

- **Experiment 1 (signal comparison):** `scripts/run_paper3_signal_comparison_hardset.sh`
- **Experiment 2 (Gate 1):** `scripts/run_paper3_gate1_real.sh`
- **Colab Pro notebook:** `notebooks/rt_paper3_signal_comparison_gate1_runner.ipynb` (downloads benchmarks from HuggingFace, persists results to Google Drive, handles LongMemEval turn-truncation for runtime management)
- **Benchmark sources:** MSC valid from `gonced8/multi-session-chat`; LongMemEval-S cleaned from `xiaowu0162/longmemeval-cleaned`; both normalized via `scripts/prepare_public_benchmark_jsonl.py`
- Frozen artifact bundles for Papers 1–3 under `artifacts/`
- Manuscript assets generated from tracked JSON under `manuscript/generated/`

The build entry point is:

```
python scripts/build_manuscript_assets.py
tectonic manuscript/paper_checkpoint.tex
```

VIII. CONCLUSION

The RT program set out to determine whether hidden-state geometry of evolving conversation trajectories is a useful signal for conversational memory compression. The answer is now complete and precise.

Paper 1 establishes the geometric substrate: conversation-state trajectories are extremely low-rank (mean rank₉₅ 1.05–1.53) and almost perfectly decoder-coupled (geometry–logit correlation 0.989–0.994) across compact model families. Paper 2 turns this into a real control law: geometry-aware retention beats uniform on constraint-critical tasks by preferentially rescuing exact support turns that downstream answers depend on (17/36 vs. 5/36 at budget 0.35). Paper 3 turns the control law into a sparse codec family and closes the geometry-versus-semantic gap with two decisive experiments: geometry scoring is necessary for constraint-critical memory because semantic scoring cannot distinguish user constraints from their assistant echoes (17/36 vs. 7/36 support-turn rescue, $p = 0.028$); and query-conditioned geometry adds substantial and significant value inside a semantic shortlist for persona-continuity memory (Δ Kendall +0.53–0.60, answer NLL improvement $p = 0.020$ at budget 0.35).

The final contribution is not a universal codec. It is the signal-conditioned KCD framework: one compression structure (keep/compress/drop), two scoring regimes matched to memory type (geometric for constraints, semantic+query-geometry for persona), and a falsification record that defines

TABLE XI
APPENDIX BRIDGE FROM THE CHECKPOINT MANUSCRIPT TO THE FULL DERIVATION LEDGER.

Layer	Status	Core evidence	Deep-dive companion
Paper 1	Frozen	Low rank + geometry–decoder coupling	ledger
Paper 2	Stable	Geometry-aware control + support-turn rescue	experiment matrix
Paper 3	Mixed	Hard-set wins; benchmark-dependent transfer	experiment matrix
Extensions	Failed / weak	MSC curvature, atlas, state-update tests	negative matrix

where the signal distinction matters and where geometry-only approaches fail. The north star of the program—not one universal geometry codec, but a signal-conditioned sparse codec family for conversational memory—is reached.

APPENDIX A

STORAGE ACCOUNTING FOR SPARSE CODECS

Any honest codec comparison must count more than just retained raw tokens. For segment j , the storage model should include the anchor state, sparse support memory, and metadata:

$$\text{Storage} = \sum_j \left(\underbrace{\|a_j\|}_{\text{anchor}} + \underbrace{\|s_j\|}_{\text{sparse support}} + \underbrace{\|\text{meta}_j\|}_{\text{segment metadata}} \right). \quad (14)$$

Future work should refine this accounting into exact byte counts for latent summaries and decoder-compatible surrogates.

APPENDIX B

RESEARCH LEDGER BRIDGE

This checkpoint is paired with a companion derivation ledger at `papers/rt_derivation_ledger.md` and a full program findings synthesis at `papers/rt_full_program_findings.md`. The ledger is the canonical deep-dive reference for the full math-to-code-to-artifact chain: every mathematical object, every benchmark family, every major experiment phase, and every explicit falsification branch. Machine-readable inventories at `papers/generated/rt_experiment_matrix.csv` and `papers/generated/rt_negative_result_matrix.csv`. Table XI gives the condensed bridge from the manuscript-level claims to the deeper ledger.

APPENDIX C

CHECKPOINT ARTIFACT MAP

The most important tracked bundles are:

- Paper 1 final: `artifacts/paper1/expanded_v8_final`

- Paper 1 regime atlas limits: artifacts/paper1/regime_atlas_smoke_v4
- Paper 2 hard stress: artifacts/paper2/behavior_stress_v1
- Paper 2 mechanism cases: artifacts/paper2/behavior_stress_qwen_cases
- Paper 3 fairness: artifacts/paper3/paper3_batch_v1_fairness
- Paper 3 3B probe: artifacts/paper3/paper3_batch_v1_3b
- Paper 3 MSC persona falsification: artifacts/paper3/msc_persona_curvature_v1
- Paper 3 state-update falsification: artifacts/paper3/state_update_alignment_smoke_qwen05b
- Paper 3 state/update cross control: artifacts/paper3/state_update_cross_control_qwen05b
- Experiment 1 signal comparison (Drive): rt_results/paper3/studies/paper3_signal_comparison_hardset_v1
- Gate 1 MSC oracle (Drive): rt_results/paper3/harm_oracle/paper3_gate1_oracle_msc_valid
- Gate 1 MSC refinement study (Drive): rt_results/paper3/studies/paper3_gate1_refinement_msc_valid

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